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Brief article

Rescuing stimuli from invisibility: Inducing a momentary release from visual masking with pre-target entrainment

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ABSTRACT

At near-threshold levels of stimulation, identical stimulus parameters can result in very different phenomenal experiences. Can we manipulate which stimuli reach consciousness? Here we show that consciousness of otherwise masked stimuli can be experimentally induced by sensory entrainment. We preceded a backward-masked stimulus with a series of rapid visual events presented at 12 Hz for 800 ms. Peaks in visual sensitivity (d') were observed when the target appeared at the time that the next entraining stimuli would have occurred. Observers' sensitivity for identical masked near-threshold stimuli increased by factors as large as 55%, but only at this precise moment in time. These data thus reveal that awareness of near-threshold stimuli can be manipulated by entrainment to rhythmic events, supporting the functional role of induced oscillations in underlying cortical excitability, and suggest a plausible mechanism of temporal attention.

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1. Introduction

When stimuli are masked to be at the threshold of consciousness, identical stimulus presentations can result in very different states of awareness. Psychophysical laws interpret these variations as stochastic phenomena and do not address why any given stimulus reaches awareness. However, given that the stimulus parameters are identical, the source of this variability in awareness must reflect variability in the state of the visual system, which is typically not under experimental control. Here we ask whether these differing states can in fact be manipulated, inducing predictable variations in awareness via rhythmic visual signals in the environment.

1.1. Entraining to rhythmic events

Prior work has shown modulation of auditory processing by a rhythmic series of tones, resulting in enhanced sample matching at times in-phase with the preceding tones (Jones, Moynihan, MacKenzie, & Puente, 2002). Jones et al. presented participants with a target tone at the beginning of a trial, followed by a series of irrelevant tones with an embedded rhythmicity. A probe tone was presented at variable intervals following the end of the rhythmic sequence. Highest matching-task performance was found on trials when the probe tones were presented in-phase with the rhythmic sequence, compared to those presented out-of-phase. The authors interpreted this finding in terms of "tuning" of temporal attention by predictable events; periodic tones (even if irrelevant) had the effect of narrowing the temporal selection window of attention (more precise time-based selection), and increasing the attentional gain (better signal-to-noise ratio). Similarly, it has been shown that oddball targets in a series of attended standard tones elicit larger mismatch negativity (MMN) in a constant rate train than those in a variable rate train of tones (Erickson & Hetrick, 2009), again indicating enhanced auditory processing for environmental stimuli occurring at predictable times.

Following Jones et al.'s work in the auditory modality, a related effect has been documented in temporal attention

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in the visual modality (the "attentional awakening" effect, see Ariga & Yokosawa, 2008; Ambinder & Lleras, 2009). In those experiments, participants were asked to identify a white letter embedded in a rapid stream of visually presented (RSVP) blue letters. Ariga and Yokosawa observed that performance in this task was poor for targets appearing early on in the RSVP stream, but improved to asymptote as the RSVP unfolded. Much like Jones et al.'s auditory temporal tuning account, this effect was interpreted as evidence of temporal attention tuning-into the timing of the series of visual events with the goal of better processing each individual item.

Moreover, it has recently been shown that sensory processing and awareness vary with respect to the phase of the ongoing EEG alpha rhythm (Busch, Dubois, & VanRullen, 2009; Lakatos, Karmos, Mehta, Ulbert, & Schroeder, 2008; Mathewson, Gratton, Fabiani, Beck, & Ro, 2009). Given that rhythmic visual stimuli have been shown to induce comparable oscillations in neural firing (Herrmann, 2001; Lakatos et al., 2008; Pastor et al., 2002; Williams, Mechler, Gordon, Shapley, & Hawken, 2004), the present study also sought to corroborate the findings of concomitant oscillations in neural firing and awareness, investigating for the first time the effect of the phase of these induced oscillations on visual awareness.

1.2. Current study

Our goal in the current study was to ask whether the visual system could be entrained to produce temporally well defined peaks in sensitivity at predictable times in a sequence. Moreover, we asked whether such entrainment could be used to modulate awareness of near-threshold masked stimuli. That is, given attention to a single location in space, can predictable events in the environment improve visual sensitivity to better process brief periods in time, such that we can manipulate which stimuli will reach consciousness? We hypothesized that the threshold of visual awareness could be entrained or synchronized to the expected occurrence of rapid, rhythmic events; with maximal sensitivity occurring for targets presented in-phase with the preceding entraining stimulation, and reduced sensitivity for out-of-phase targets. The experimental task was to detect the presence of a luminance disk at fixation. We used a metacontrast mask (at optimal stimulus-onset-asynchrony; SOA) to reduce target visibility to near-threshold levels and minimize contributions from trace processing of post-target visible persistence activation by diminishing re-entrant activity (Di Lollo, Enns, & Rensick, 2000; Enns, 2004; Ro, Breitmeyer, Burton, Singhal, & Lane, 2003).

2. Method

2.1. Participants

Sixteen students from the University of Illinois at Urbana-Champaign received course credit in a Psychology class for participation and signed informed consent.

2.2. Materials and design

Subjects were seated 57 cm away from a 21 in CRT monitor set to 85 Hz in a dimly lit room. Fig. 1 shows the stimulus dimensions and task sequence. All stimuli were presented at fixation. The target consisted of a 1° disk. The mask annulus was 2°, with a 1° center cut out, and was also used as the entraining stimulus. All stimulus presentations were multiples of the 11.7 CRT monitor refresh cycle, thus representing single or double pulsed presentations.

After a 247 ms fixation cross and a 400 ms blank screen, a variable number of masks were presented as entrainers (0, 2, 4, or 8). Entrainers were presented at 12.1 Hz for 23.6 ms each (two monitor refreshes), yielding an entrainer-stimulus onset asynchrony of 82 ms (eSOA). We varied five different SOAs between the final entrainer and the target onset (tSOA): a tSOA of 82 ms (when the target appeared in-phase with the entraining sequence); as well as out-of-phase tSOAs ahead of (32 ms and 59 m), and after (107 and 130 ms) the expected time. After the tSOA, the target (presented for 11.7-ms; one monitor refresh) was followed by a metacontrast mask annulus (presented for 23.6-ms; two monitor refreshes) with a constant 45-ms target-mask SOA (mSOA). The subject then had 1500 ms to respond before the next trial began, with a fixation cross reappearing 300 ms after the appearance of the final mask. On 20% of trials, the target was omitted in order to assess false alarms.

A control condition was included with only two pre-target annulus stimuli separated by a 574 ms interval in order to simultaneously control for the forward masking effects of the entrainers (Breitmeyer, 1984), the foreperiod length of the 8-entrainer condition (Bausenhart, Rolke, & Ulrich, 2008; Drazin, 1961), and the overall number of stimuli in the 2-entrainer condition.

2.3. Procedure

Subjects were instructed to indicate on each trial whether or not they detected the presence of the target, which they were told was absent on some trials. They did not receive feedback on their detection performance. After completing a practice block, subjects completed 12 blocks of trials with self-paced rest periods in between each block. Each block consisted of 75 trials, randomly intermixing 15 of each entrainment condition (0, 2, 4, 8, or control). The tSOA (32, 59, 82, 107, or 130 ms) was also randomly varied throughout each block.

3. Results

The average detection rate in the no-entrainers condition ($M = 0.41$) indicated considerable metacontrast masking. It should be noted, however, that this condition is not directly comparable to the remaining conditions because it lacks a forward masking stimulus: that is to say, in all entraining conditions (as well as in the control condition), the last entraining annulus in the sequence served as a for-

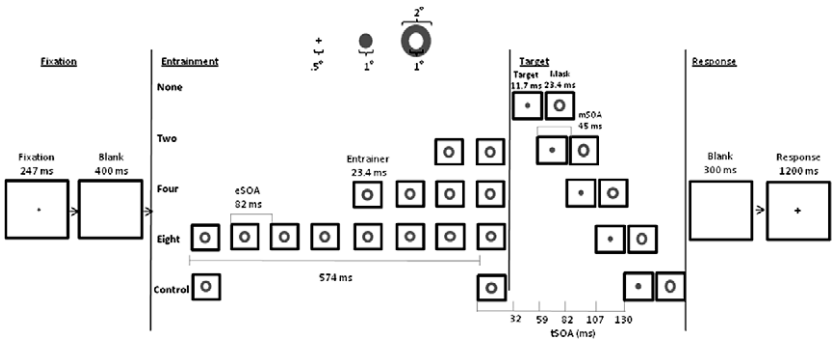


Fig. 1. Stimulus dimensions and trial timeline.

ward masking stimulus to the target, further reducing target visibility.

Fig. 2 plots the average detection rate for each entrainment condition and the control condition as a function of tSOA between the final entrainer (or control stimulus) and the target. Target detection in the control condition was lower ($M = 0.22$) than in the no-entrainer condition, confirming the forward masking effects of the last "entrainer".

Both linear and quadratic trends as a function of tSOA were computed with orthogonal linear contrasts weights, separately for each entrainment condition. There was a small and positive linear trend in the control condition, with detection performance increasing slightly with increasing tSOA ($t(15) = 3.0$, $p < .01$). This result suggests the possibility of decreased forward masking at longer tSOA intervals, or warning effects arising at the longer tSOAs. That said, there were no significant linear trends observed in the 2-, 4- and 8-entrainers conditions (all $t < 2$). Moreover, the difference in linear trends between the control condition and the entrainment conditions was confirmed by a fixed-effects ANOVA ($F(4,60) = 3.75$, $p < .01$).

Crucially, the presentation of entraining stimuli prior to the onset of the target, elicited a peak in detection at the moment the next entrainer would have occurred (82 ms tSOA), as well as poorer detection for targets before or after this point in time. The quadratic trend across tSOAs varied with entrainment condition ($F(4,60) = 16.02$, $p < .0001$). Further, the slope of these quadratic effects across entrainment conditions was greater than zero (0-Control-2-4-8; $M_b = .22$, $t(15) = 6.53$, $p < .0001$) indicating a scaling of the effect with the number of entrainers. Planned comparisons showed that the quadratic effects for *Two* ($t(15) = 3.5$, $p < .01$), *Four* ($t(15) = 4.9$, $p < .001$), and *Eight* ($t(15) = 5.5$, $p < .0001$) entrainers were reliable, whereas the quadratic components for no and control entrainment were not different from zero ($t < 1$). The quadratic nature of this release from masking confirms our prediction of a temporally-precise effect of sensory entrainment.

Because detection rates are known to be susceptible to differences in decision biases, we also computed *d*-prime (d') for each subject, as a measure of sensitivity (Fig. 3). However, because of the low occurrence of false alarms, we collapsed observations across the 2-, 4-, and 8-entrainer

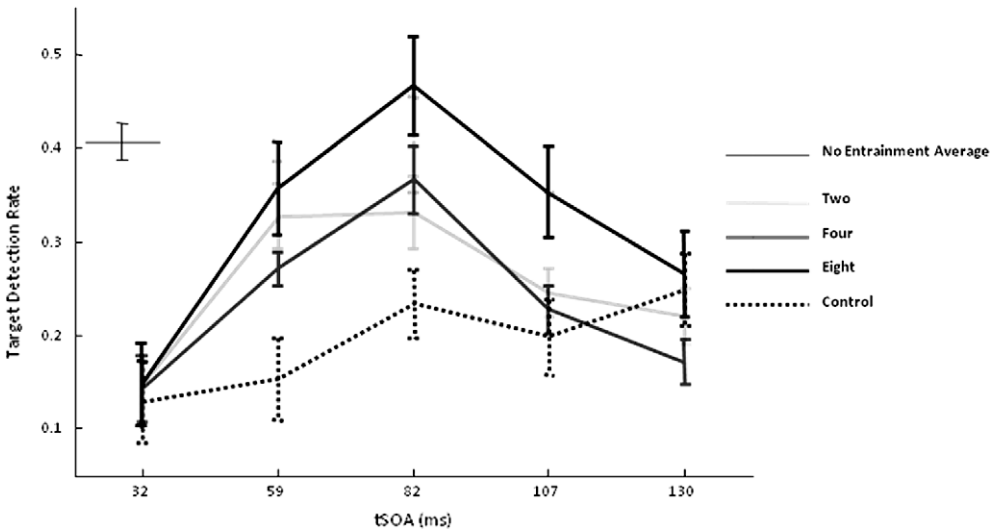


Fig. 2. Mean target detection rate as a function of the time after final entrainer onset (tSOA) for each entrainment condition across subjects. Apparent is a peak in detection in-phase with the preceding entrainment. Error bars represent within-subject SE (Loftus & Masson, 1994).

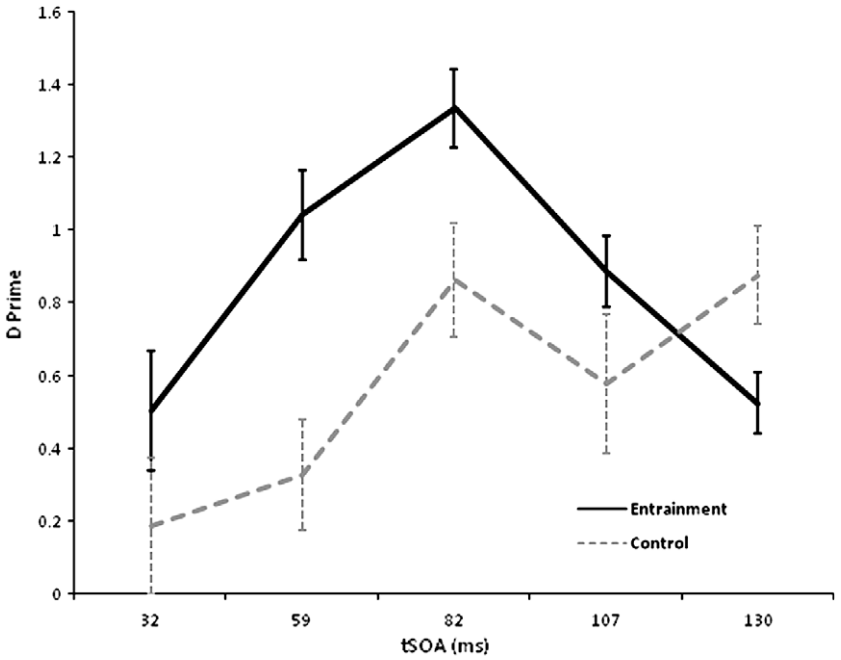


Fig. 3. Mean d' collapsed across entrainment condition compared to the control condition. Error bars represent within-subject SE (Loftus & Masson, 1994).

conditions to obtain valid estimates of d'. As expected, d' in the entrainment conditions showed a significant quadratic trend in visual sensitivity as a function of tSOA ($t(15) = 4.3$, $p < .001$), with the best sensitivity arising in-phase with the preceding entrainers. In contrast, the quadratic component in the control condition was not significant ($t(15) = .64$, *n.s.*).

Finally, planned comparisons revealed that visual sensitivity at the tSOA in-phase condition (82 ms) was significantly greater following entrainment ($M = 1.34$) than following the control condition ($M = .86$; $t(15) = 3.5$, $p < .01$). In contrast, visual sensitivity at counter-phase time points did not differ between the entrainment and control conditions (for tSOA = 35 ms: $M = .50$ vs. $M = .18$; $t(15) = 1.9$, *n.s.*; for tSOA = 130 ms: $M = .52$ vs. $M = .87$; $t(15) = 1.7$, *n.s.*), indicating that the entrainment manipulation specifically caused an increase in sensitivity at the predicted point in time (when target appeared in-phase with the entrainment sequence) rather than a decrease for targets appearing out-of-phase with the preceding entrainment. It should be further emphasized that the 8-entrainer condition resulted in a substantial (55%) increase in visual sensitivity and a 100% increase in the average detection rate compared to the control condition.

4. Discussion

Here we showed for the first time that a momentary release from the effects of visual masking can be elicited by rapid presentation of visual stimuli, causing a temporally-precise peak in visual sensitivity. When targets were presented in-phase with preceding entrainers, target detection and sensitivity were substantially improved. In contrast, targets presented out-of-phase with the previous entrainment were less likely to reach awareness, although they were no less detectable than targets in the control condition (with no entrainment).

The current findings indicate that rhythmic stimuli in the environment can influence the visual system, increasing the sensitivity to events occurring in-phase with other ongoing sensory events. One possibility is that the time at which future events are likely to occur is being *predicted* from preceding rhythmic environmental events, so as to enhance neural processing at that precise moment. In this respect the current results fit well with emerging evidence regarding the inherently predictive nature of brain processing (Enns & Lleras, 2008).

One might argue that the observed momentary release from masking merely represented a phasic shift in sensitivity occurring after any pre-target stimulus, which scaled monotonically with the number of stimuli presented. Our control condition calls into question this alternative explanation; both it and the 2-entrainer condition had an identical number of pre-target stimuli, yet the momentary spike in sensitivity at 82 ms only occurred after the rapid entrainment, when the next entrainer would have occurred. Such an account could be rescued, however, if the phasic shift in sensitivity induced by a preceding stimulus decays with time and can be summed across stimuli. Under such conditions the 2-entrainer condition would produce a higher peak in sensitivity than the control condition because the two entrainers follow each other more closely in time than the two control stimuli. Work is currently underway to distinguish between these two possible explanations for this effect.

The use of the mask as the entraining stimulus raises the possibility that the momentary increase in visual

sensitivity was an effect of changes in mask processing due to the prior presentation of identical stimuli, rather than changes in target processing. The quadratic nature of the induced oscillation calls into question any coarse adaptation account (i.e., more masks make later masks weaker). Further, such accounts would predict a different relation between masking strength and tSOA: if entrainment caused increased inhibition of mask processing at in-phase intervals after the last entrainment, then one ought to have observed the weakest masking at the tSOAs of 32 ms and 130 ms, when the mask appeared in-phase with the entrainers (entrainer-to-mask SOAs of 77 ms and 175 ms, respectively). In fact, sensitivity was worst at these tSOAs, strongly suggesting that our results do not arise from suppressed mask processing, but rather enhanced target processing.

It has recently been shown that the variability in conscious awareness may be a function of the phase of natural rhythmic alpha oscillations in the brain (Busch et al., 2009; Mathewson et al., 2009), supporting the theory that neural oscillations represent cycles of excitability in particular cortical areas (Lindsley, 1952). When presented in the trough of the oscillation, targets were less likely to be detected compared to targets presented at the peak. Both sets of results suggest that awareness of the target is strongly dependent on how ready the visual system is at any given point in time to process the incoming target. One possibility is that the induced activity elicited by the entraining stimuli measured with EEG (Herrmann, 2001; Lakatos et al., 2008; Pastor, Artieda, Arbizu, Valencia, & Masdeu, 2003; Pastor et al., 2002; Williams et al., 2004) and the endogenous oscillatory activity (Busch et al., 2009; Mathewson et al., 2009) reflect similar oscillations in cortical excitability, representing analogous mechanisms. Further supporting this claim is a recent study showing that inducing gamma activity by 50 Hz flicker has the same attentional enhancement effect as increases in naturally occurring gamma activity (Bauer, Cheadle, Parton, Muller, & Usher, 2009).

Other recent work suggests that entrainment of neural oscillations may play a part in speech perception (Schmidt-Kassow & Kotz, 2008; Schroeder, Lakatos, Kajikawa, Partan, & Puce, 2008), and attentional selection (Lakatos et al., 2008; Large & Jones, 1999). Entrained excitability cycles can begin to explain many seemingly unrelated phenomena, complementing research on the effect of ongoing oscillations on cognition (Ward, 2003). For instance, in a number of RSVP paradigms, distracters and targets are presented serially at 10 Hz (see for instance the large literature on the Attentional Blink; Olivers, 2007; Raymond, Shapiro, & Arnell, 1992), and as documented by Ariga and Yokosawa (2008), target identification rates increase as a function of the number of preceding distracters (see also Ambinder & Lleras, 2009). It may very well be that the degree of synchronization to the RSVP stream is related to the magnitude of the Attentional Blink (AB), or to the likelihood of observing a release from the AB (e.g., Arend, Johnston, & Shapiro, 2006; Olivers & Van der Burg, 2008). This conjecture is consistent with results from an MEG study showing evidence of cortical synchronization to RSVP stimulation (Gross et al., 2004).

5. Conclusion

Our results show that in the presence of rapid, periodic visual events, visual processing appears exogenously synchronized to the frequency of visual stimulation to produce phase-locked peaks of sensitivity. This entrainment may represent a mechanism by which temporal attention is tuned to produce temporally-precise peaks in visual sensitivity, to both anticipate and optimize visual processing of brief visual events. The phase effects here mirror those related to endogenously generated brain rhythms, raising the possibility that the two phenomena rely on similar underlying mechanisms. Rhythmic visual entrainment has a dramatic effect on visual sensitivity, providing us with a powerful technique to experimentally control with fine temporal precision whether or not near-threshold stimuli reach conscious awareness.

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